WELLSLING · TRACEABILITY MATRIX

WellSling traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wellsling		
Source	docs/03b-traceability.md @ 346fa2d		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.3	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.2	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

WellSling traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 17 user requirements, 17 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	No entry	WSL-CAL-001, WSL-BLD-001	Inspection of the method (BLD-001), TRL 3; trial with a weighted dummy in a test pit, TRL 4	Met by design
UR-002, UR-011	PR-002	Lifting capacity	WSL-CAL-001, WSL-BLD-001	Analysis [A2, B1, B10, C5, C6], TRL 3; proof-load test, TRL 4	Met on paper (static; the dynamic load of a kicking animal to measure at TRL 4)
UR-008	PR-003	Band spreads the load	WSL-CAL-001	Analysis [E6, E7], TRL 3; design review with a veterinary partner, TRL 4	Met on paper for width: two 240 mm bands, about 17 to 21 kPa; tolerance to verify at TRL 4
UR-003	PR-004	Passer reach	WSL-CAL-001, WSL-BLD-001	Analysis [G3], TRL 3; trial in a test pit, TRL 4	Met on paper: 16.5 m
UR-004	PR-005	Well span	WSL-CAL-001, WSL-BLD-001	Analysis [C9, F4], TRL 3; fit check on sample wells, TRL 4	Met: pads 700 mm, bearers 400 mm back
UR-005	PR-006	Fast setup	WSL-CAL-001, WSL-BLD-001, WSL-STM-001	Analysis [G4], TRL 3; timed drill, TRL 4	At risk: 20 min estimate with no margin
UR-007	PR-007	Controlled lift	WSL-CAL-001, WSL-BLD-001	Inspection of the winch specified, TRL 3; brake hold test at proof load, TRL 4	Met by selection of a load-pressure brake
UR-006	PR-008	Simple operation	WSL-CAL-001, WSL-BLD-001	User trial with first-time operators, TRL 4	To verify at TRL 4
UR-014	PR-009	Value-engineering target	WSL-CAL-001	Costed BOM [G2], TRL 3	Over target: USD 3,093, USD 1,593 over
UR-011	PR-010	Tripod stability	WSL-CAL-001, WSL-BLD-001	Analysis [C3, C4], TRL 3; lean test at proof load, TRL 4	Met on paper: tips only at 19.4 deg
UR-010	PR-011	Hand effort	WSL-CAL-001, WSL-BLD-001, WSL-STM-001	Analysis [B6], TRL 3; force gauge on the handle, TRL 4	Met on paper: 200 N at most, on an assumed handle force
UR-009	PR-012	Landing clearance	WSL-CAL-001, WSL-BLD-001	Analysis [F5], TRL 3; fit check with a dummy, TRL 4	Met on paper: 169 mm
UR-013	PR-013	Carry weight	WSL-CAL-001, WSL-BLD-001, WSL-STM-001	Analysis [G1], TRL 3; weigh the parts, TRL 4	Met on paper: 32.0 kg
UR-012	PR-014	Rim restraint		Inspection of the kit and method (BOM line 28,	Met by design

User requirement	Product requirement	Title	Evidence	Verification	Status
			WSL-CAL-001, WSL-BLD-001	BLD-001), TRL 3; drill, TRL 4	
UR-015	PR-015	Outdoor life	WSL-CAL-001	Inspection of the BOM and analysis (STEM S28), TRL 3; coating gauge, TRL 4	Met by design
UR-016	PR-016	Local making	WSL-CAL-001, WSL-BLD-001	Inspection of the drawings and build plan (BLD-001), TRL 3	Met by design
UR-017	PR-017	Design-arounds	WSL-CAL-001	Inspection of the BOM (lines 18 to 22), TRL 3	Met by design

Gaps

None.